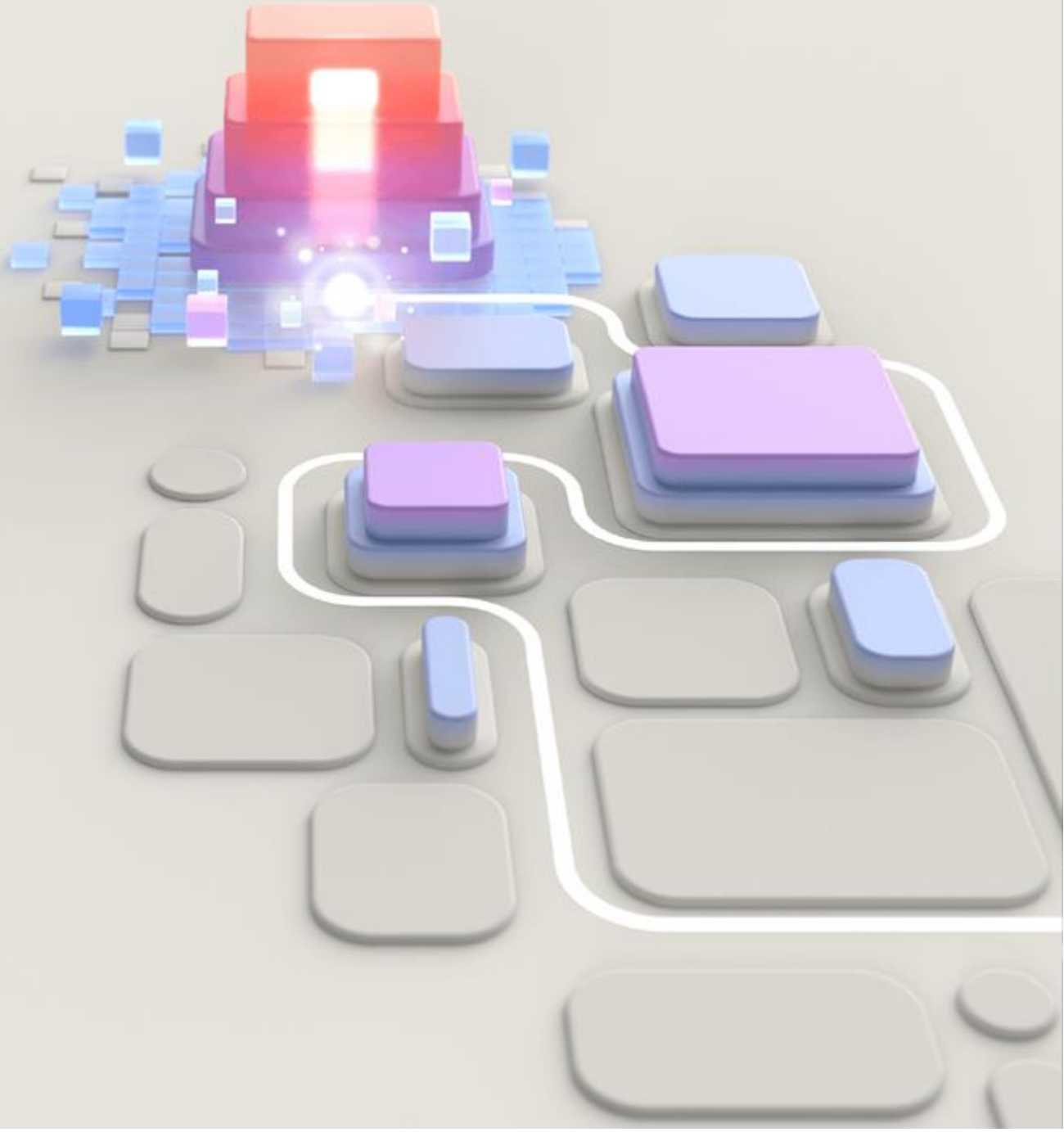
Microsoft

AZ-104

Administer Virtual Networking



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AZ-104 Agenda

01: Administer Identity
02: Administer Governance and Compliance
03: Administer Azure Resources
04: Administer Virtual Networking
05: Administer Intersite Connectivity
06: Administer Network Traffic Management
07: Administer Azure Storage
08: Administer Azure Virtual Machines
09: Administer PaaS Compute Options
10: Administer Data Protection
11: Administer Monitoring

Learning Objectives - Administer Virtual Networking

- Configure Virtual Networks
- Configure Network Security Groups
- Host your domain on Azure DNS
- Lab 04 – Implement Virtual Networks

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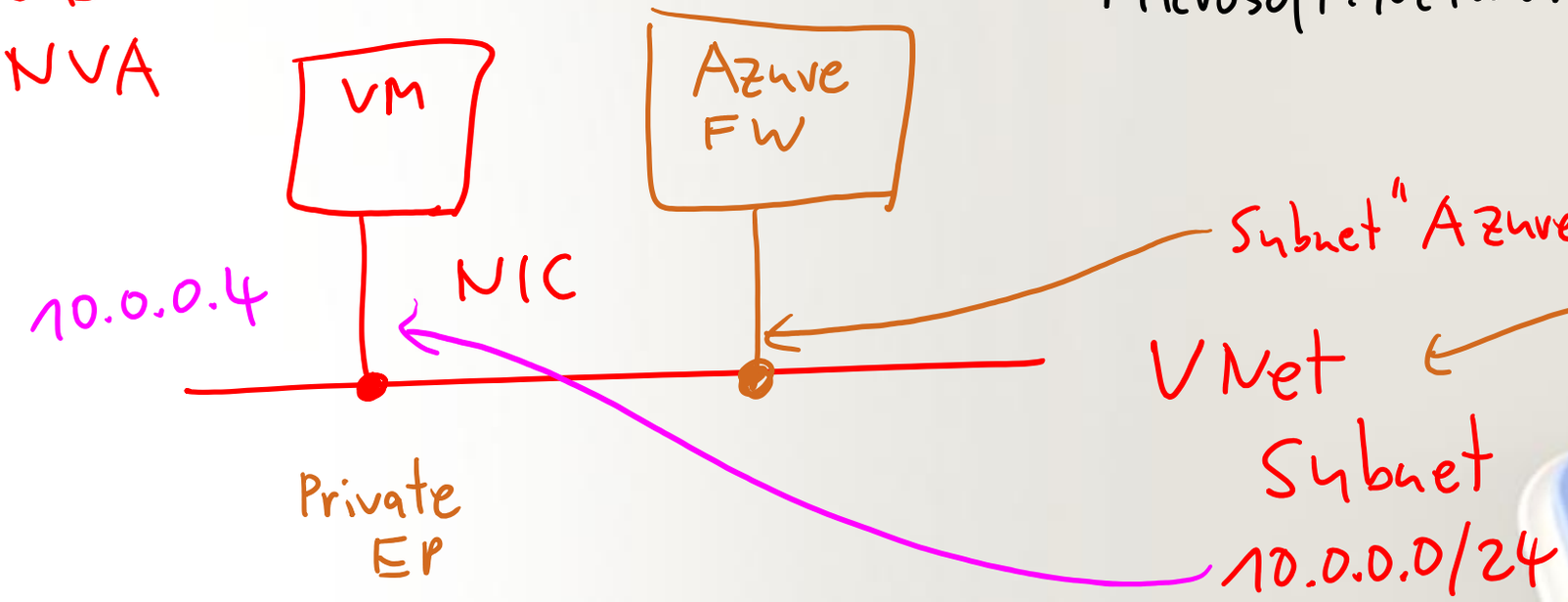
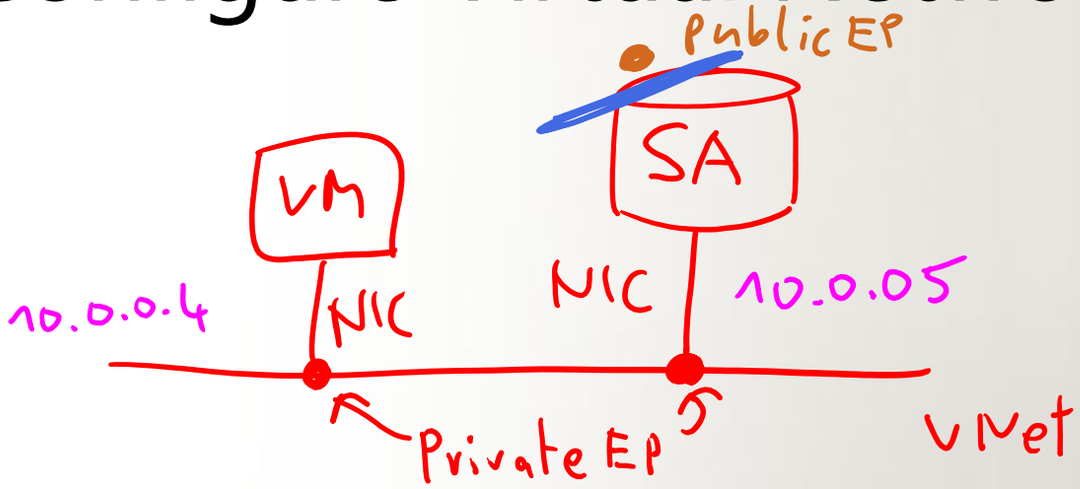
UDR (BGP geht auch)

NVA



Microsoft.Network/virtualNetworks/subnet/
parent / child

Configure Virtual Networks



192.168/16
172.16/12

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Learning Objective - Configure Virtual Networks

- Review Virtual Network Fundamentals
- Review IP Addressing Fundamentals
- Plan for IP Addressing
- Review Examples of Public IP Addressing
- Demonstration – Virtual Networks and IP addressing
- Learning Recap

Exam objectives

Configure and manage virtual networks in Azure

- Create and configure virtual networks and subnets
- Configure public IP addresses

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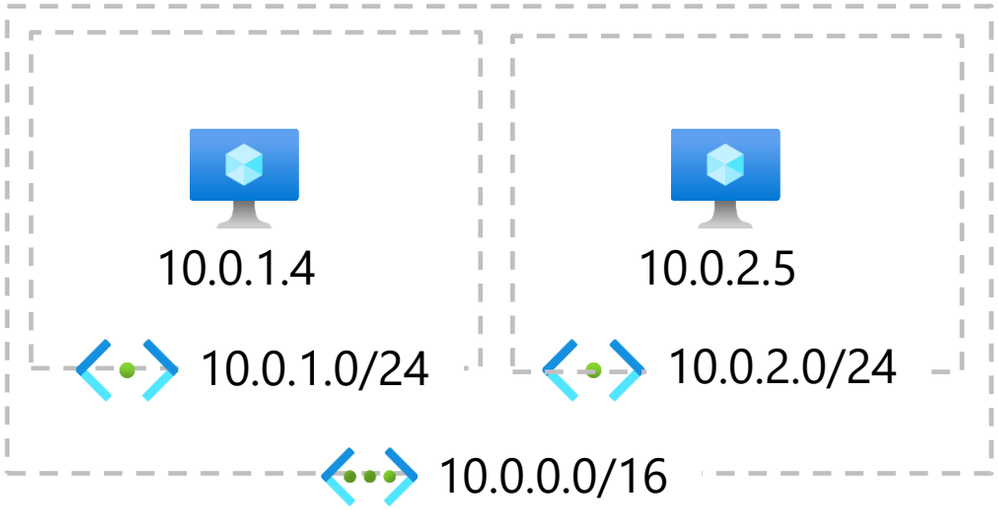
Review Virtual Network Fundamentals

Vint Cerf
1963
TCP IP 32 Bit ~ $4 \cdot 10^9$

- **Virtual Network (VNet):** The main boundary, a large private IP address space (e.g., 10.0.0.0/16) representing your isolated cloud network.
- **Subnets:** Smaller divisions within the VNet (e.g., 10.0.1.0/24, 10.0.2.0/24) for grouping resources, often public (with gateways) or private (for databases).
- **IP Addresses:** Individual addresses assigned to resources (VMs, databases) within subnets, following the subnet's range (e.g., 10.0.1.4, 10.0.2.5).
- **CIDR Notation:** /16, /24 notation showing network size and available IP addresses.



IPv6 2001:cafe::/64

Pause bis 11^{00}

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Review IP Addressing Fundamentals

- Not all IP addresses are available for assignment.
- The first four and the last IP address in every subnet is reserved for internal platform use (network/broadcast, default gateway, Azure DNS).
- Multicast, Broadcast, Loopback, Link-Local, and Internal DNS IP address ranges are not available for assignment.
- RFC 1918 defines specific IPv4 address ranges for private networks. These addresses are not routed on the public internet.

10.0.0.0/16

10.0.0.0	/16
10.0.0.0 - 10.0.255.255	65,536 addresses

Subnets	IP address range	Size
default	10.0.0.0 - 10.0.0.255	/24 (256 addresses)

.1 default GW

.2
.3 } DNS reserviert

.4

⋮

254

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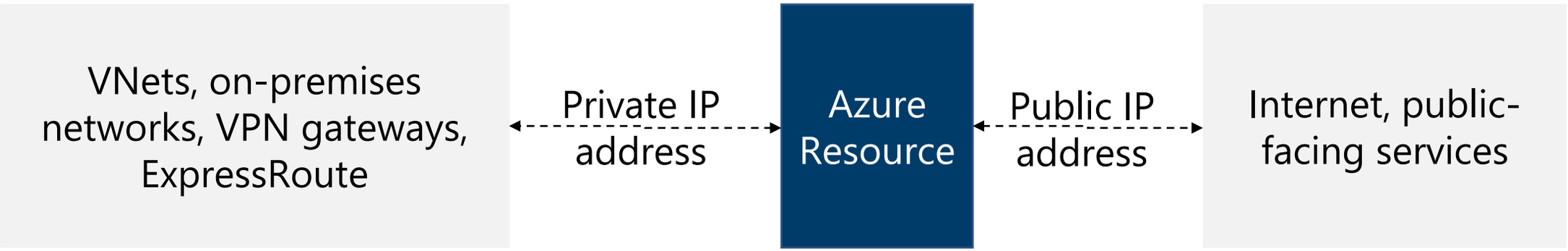
Plan for Azure IP Addressing



- Ensure address spaces don't overlap
- Use a few large virtual networks instead of multiple smaller ones
- Decide if your IP addresses should be private or public
- Decide if the IP addresses should be statically or dynamically allocated – consider using separate subnets
- Subnets shouldn't cover the entire address space of the virtual network
- Never manage IP addresses from within the virtual machine

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Review Examples of Public IP Addressing

- Public IP addresses allow Internet resources to communicate inbound to Azure resources
- Enables communication to the internet and public-facing Azure services
- Dedicated exclusively to the resource until you unassign it or delete the resource
- Public IP address prefixes (a reserved range of public IP addresses in Azure) are also available.

Public IP addresses	IP address association
Virtual Machine	NIC
Load Balancer	Front-end configuration
VPN Gateway	Gateway IP configuration
Application Gateway	Front-end configuration
Azure Firewall	Front-end configuration

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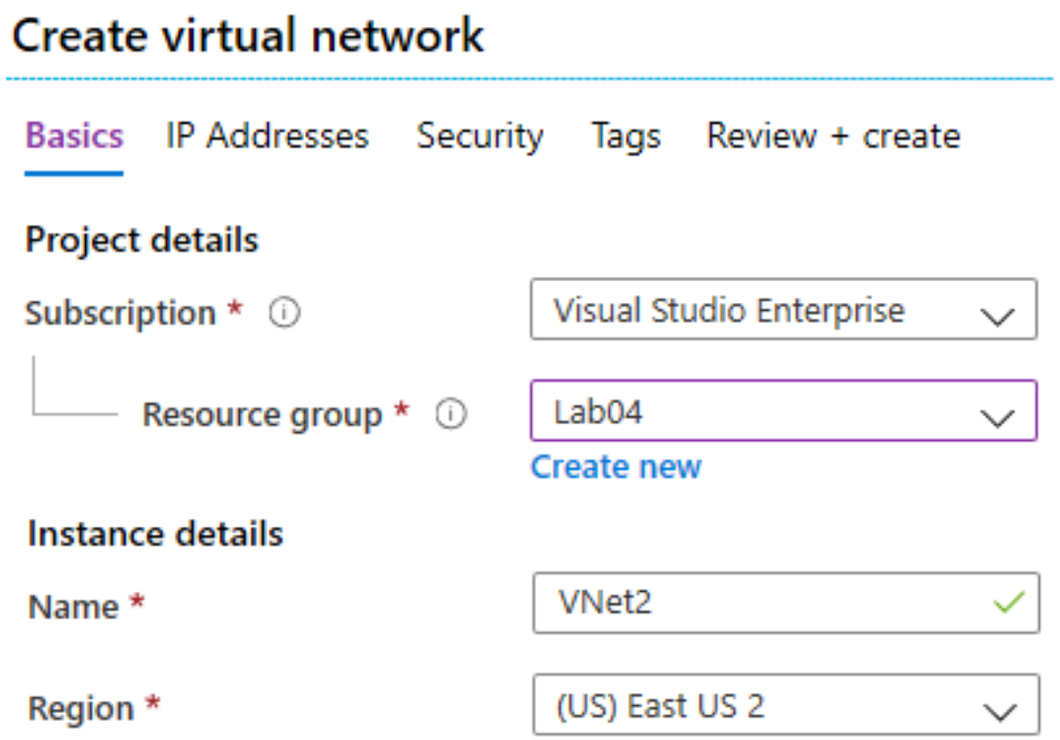
Demonstration – Virtual Networks and IP Addressing

- Create a virtual network in the Azure portal
- Configure subnets
- Create a public IP address



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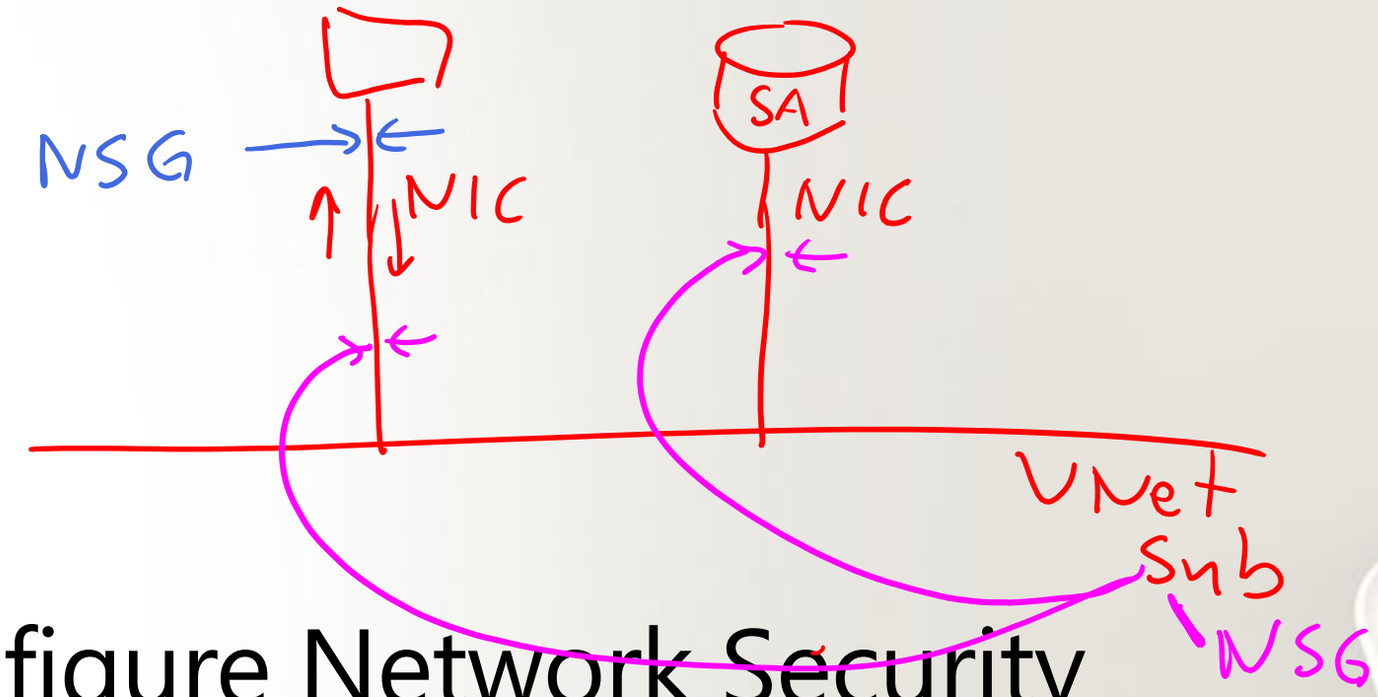

Configure Network Security Groups (NSGs)

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Learning Objectives – Network Security Groups

- Implement Network Security Groups
- Determine NSG Rules
- Determine NSG Effective Rules
- Create NSG Rules
- Implement Application Security Groups
- Demonstration – NSGs
- Learning Recap

Exam objectives

Configure secure access to virtual networks

- Create and configure network security groups and application security groups
- Evaluate effective security rules in NSGs

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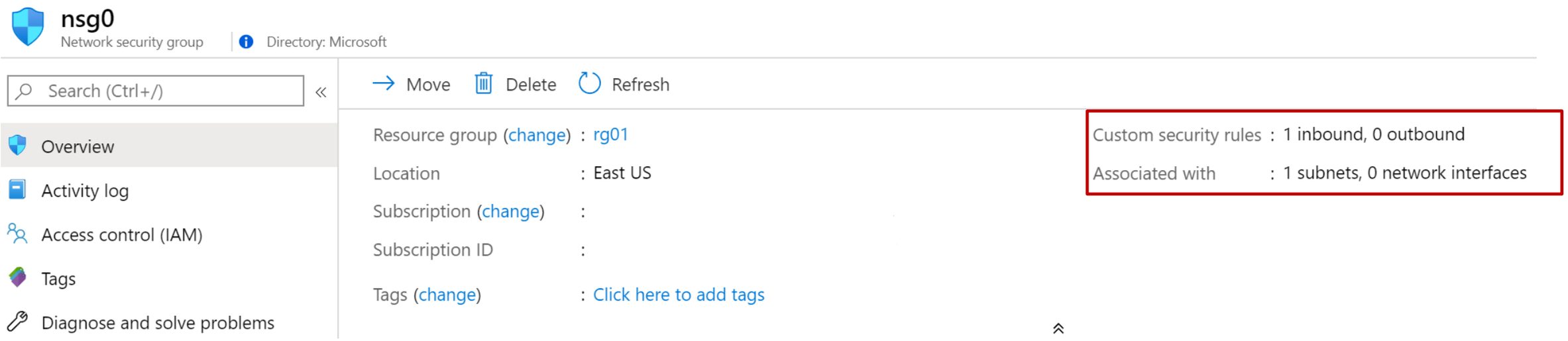
Implement Network Security Groups



- Limits network traffic to resources in a virtual network
- Lists the security rules that allow or deny inbound or outbound network traffic
- Associated to a subnet or a network interface
- Can be associated multiple times

NSG — NIC1 - VM1
NIC2 - VM2

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Determine NSG Rules

Inbound security rules

Priority	Name	Port	Protocol	Source	Destination	Action
65000	AllowVnetInBound	Any	Any	VirtualNetwork	VirtualNetwork	Allow
65001	AllowAzureLoadBalancerInBound	Any	Any	AzureLoadBalancer	Any	Allow
65500	DenyAllInBound	Any	Any	Any	Any	Deny

Outbound security rules

Priority	Name	Port	Protocol	Source	Destination	Action
65000	AllowVnetOutBound	Any	Any	VirtualNetwork	VirtualNetwork	Allow
65001	AllowInternetOutBound	Any	Any	Any	Internet	Allow
65500	DenyAllOutBound	Any	Any	Any	Any	Deny

- Security rules in NSGs enable you to filter network traffic that can flow in and out of virtual network subnets and network interfaces
- There are default security rules. You cannot delete the default rules, but you can add other rules with a higher priority

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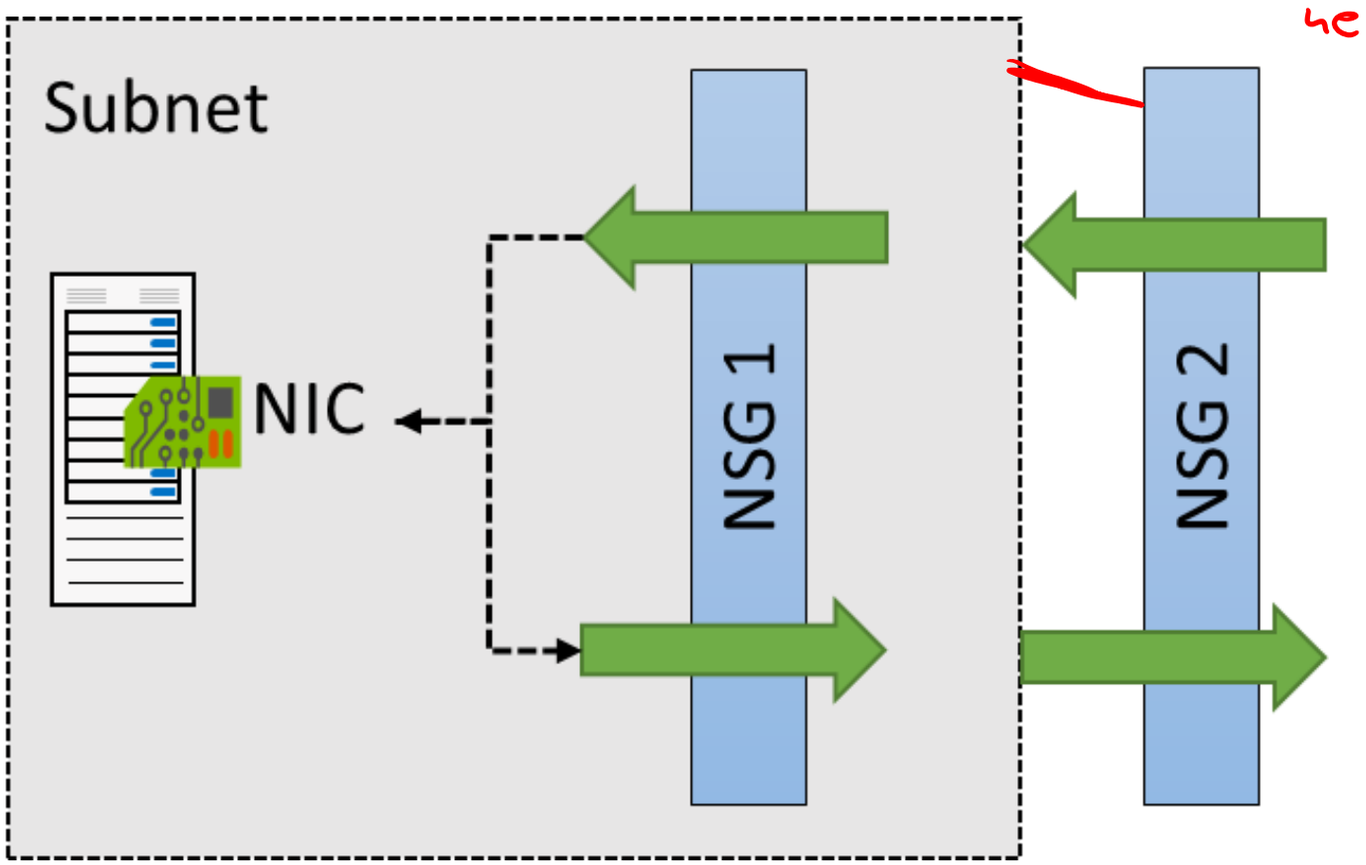
Determine NSG Effective Rules

- NSGs are evaluated independently for the subnet and NIC
- An “allow” rule must exist at both levels for traffic to be admitted
- Use the Effective Rules link if you are not sure which security rules are being applied



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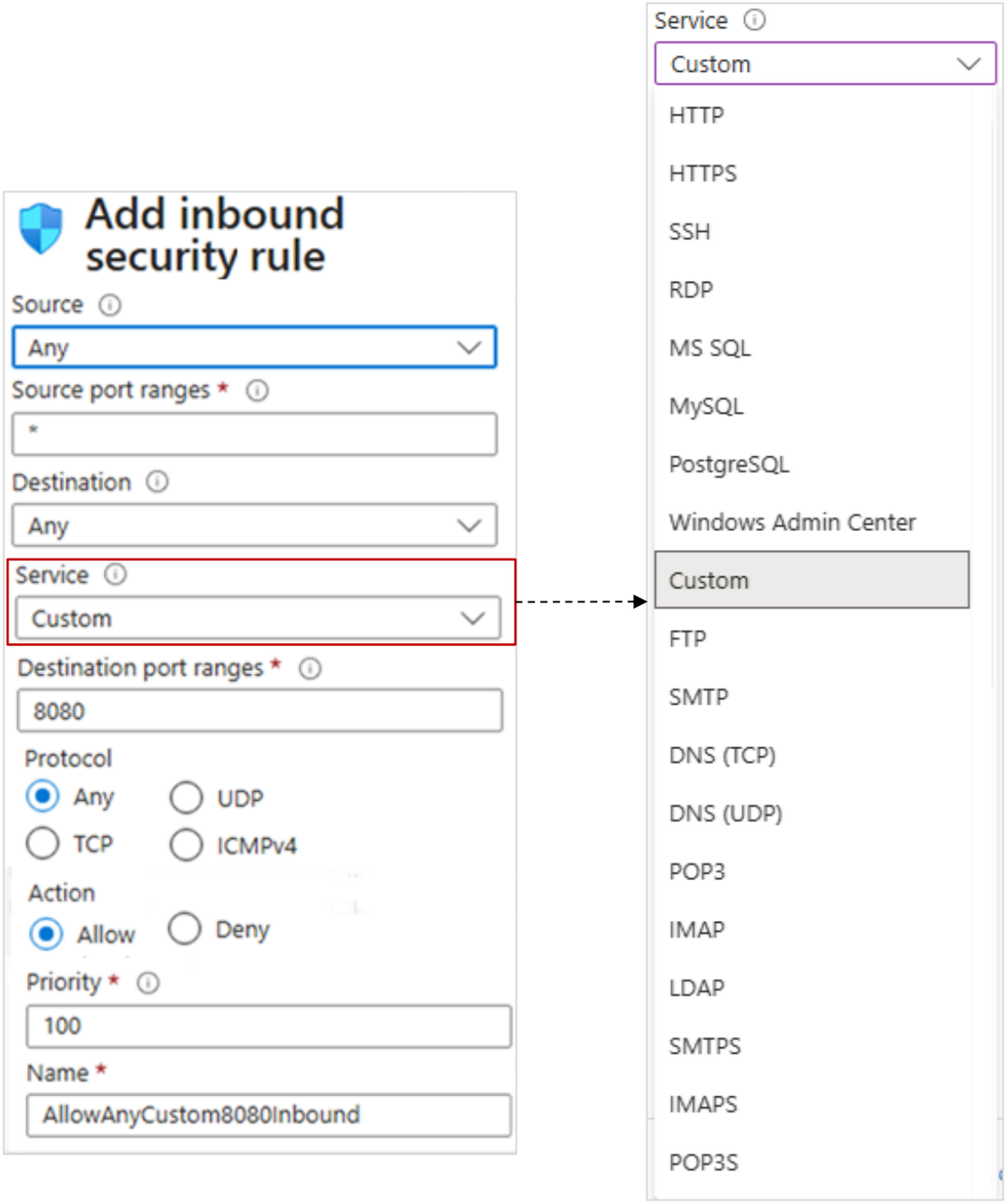
Create NSG rules

- For each rule, you can specify source and destination details, port, and protocol.
- Select from many preconfigured services, or create your own custom rules
- Rules are processed in priority order. The lower the number, the higher the priority.
- There are limits on the number of rules per NSG for each Azure location and subscription.



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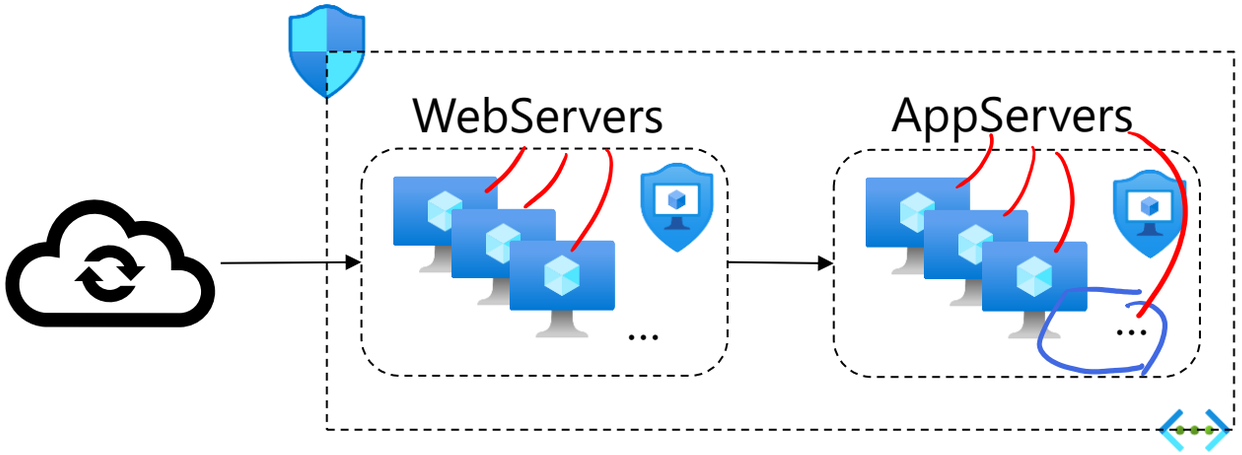
Implement Application Security Groups (ASG)

- Group VMs by application or workload and apply network security policies to those groups - instead of to individual IP addresses.
- ASGs must be in the same region and subscription as the VMs they group.
- ASGs can be the source or destination in a security rule



Action	Source	Destination	Port
Allow	Internet	WebServers	80,443
Allow	WebServers	AppServers	443
Deny	Any	Any	Any

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Host your domain on Azure DNS

(Public) DNS Zone

nslookup www.contoso.com

Resolve-DNSName

SOA
NS 171.0.8.15
www A 20.1.2.3
AAAA
CNAME

contoso.com

www.contoso.com → 20.1.2.3

Private DNS Zone

nslookup vm2

resolve
register
Link

vm1

vm2

contoso.local

VNet
Sub

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Learning Objective - Host your domain on Azure DNS

- Review DNS Fundamentals
- What is Azure DNS?
- Create a Public DNS Zone and a DNS Record Set
- Configure a Private DNS zone
- Demonstration – Azure DNS
- Learning Recap

Exam objectives

Configure name resolution and load balancing

- Configure Azure DNS

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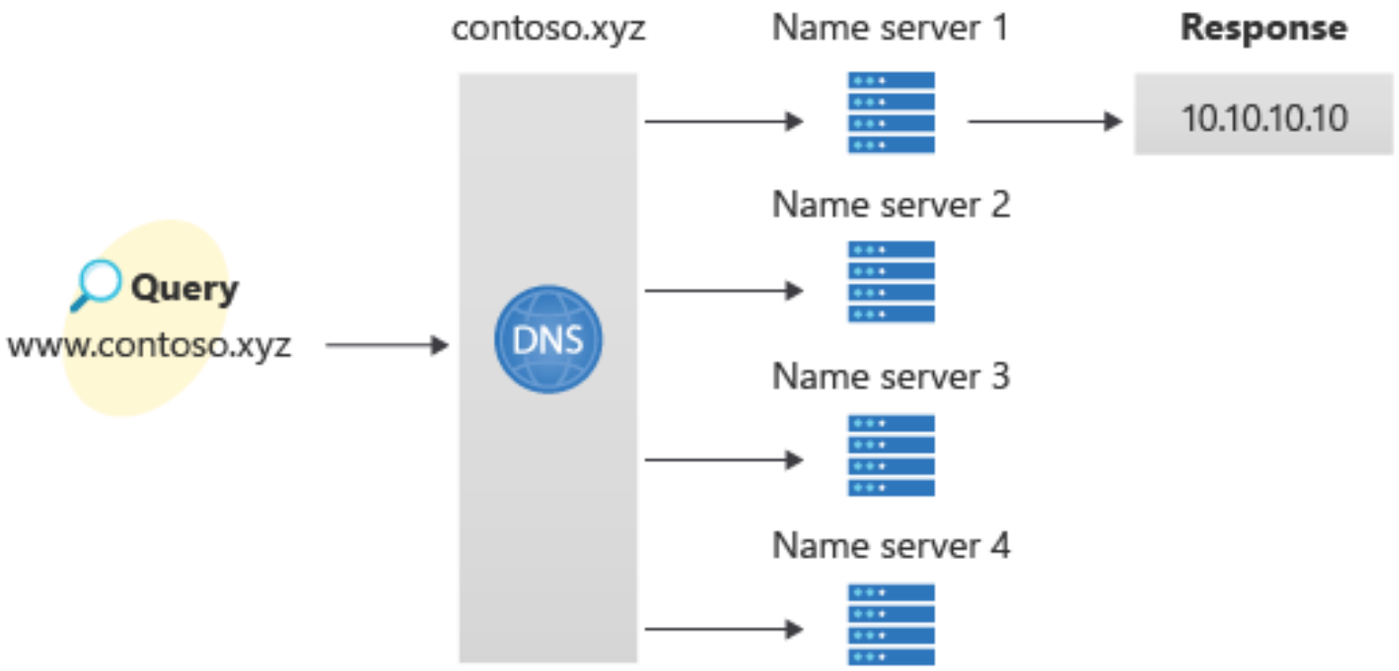
Review Domain Name Service (DNS) Fundamentals

- **DNS Zone:** Hosts the DNS records for a particular domain.
- **DNS Records:** Information to handle requests.
  - A: Maps a domain name to IPv4 address
  - AAAA: Maps a domain name to IPv6 address
  - CNAME : Maps for one domain to another (alias record)
- **DNS Resolution:** The process of translating a human-readable domain name into a machine-readable IP address.



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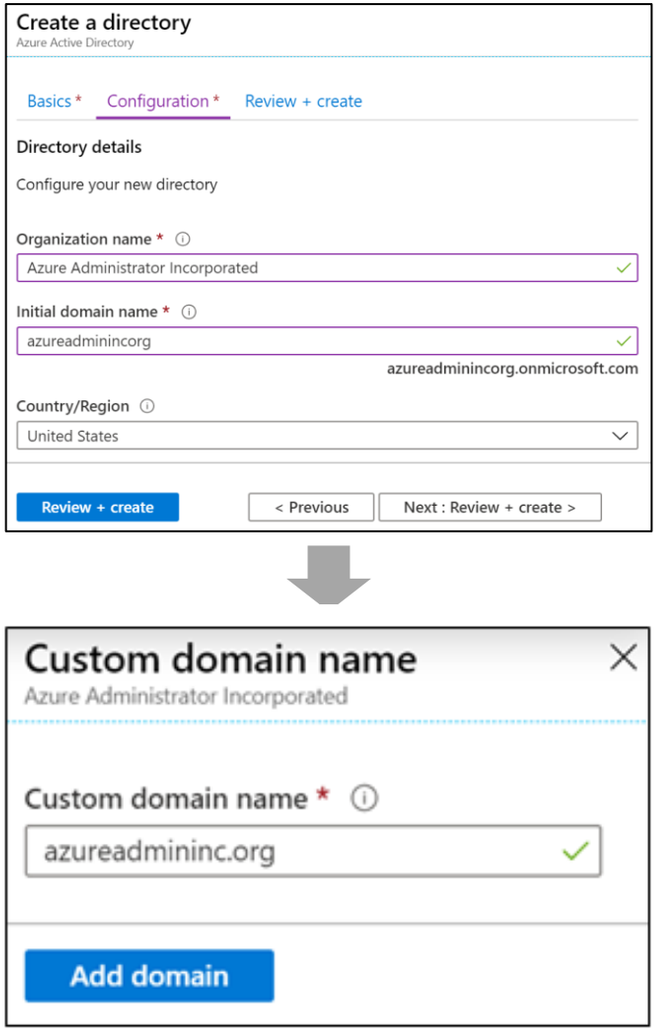
What is Azure DNS?

Tenant

- When you create a new tenant, a new default domain is created
- The domain has initial domain name in the form *domainname.onmicrosoft.com*
- You can add a custom domain name
- After the custom name is added it must be verified – this demonstrates ownership of the domain



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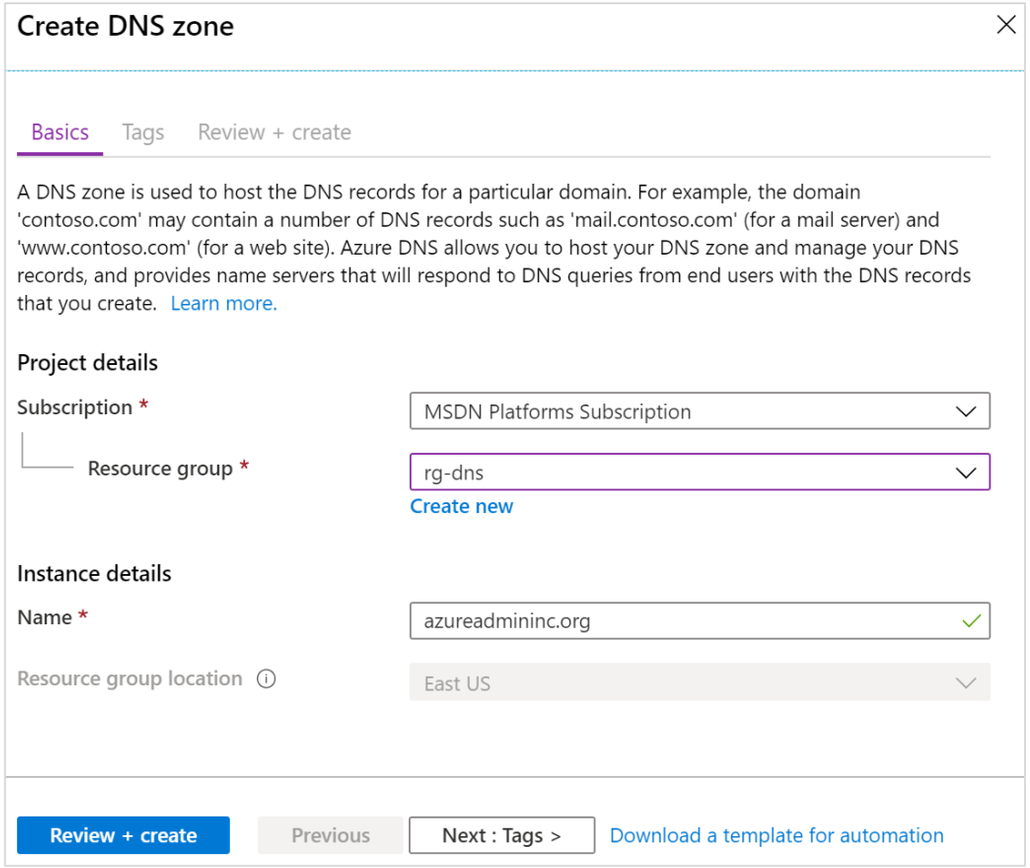
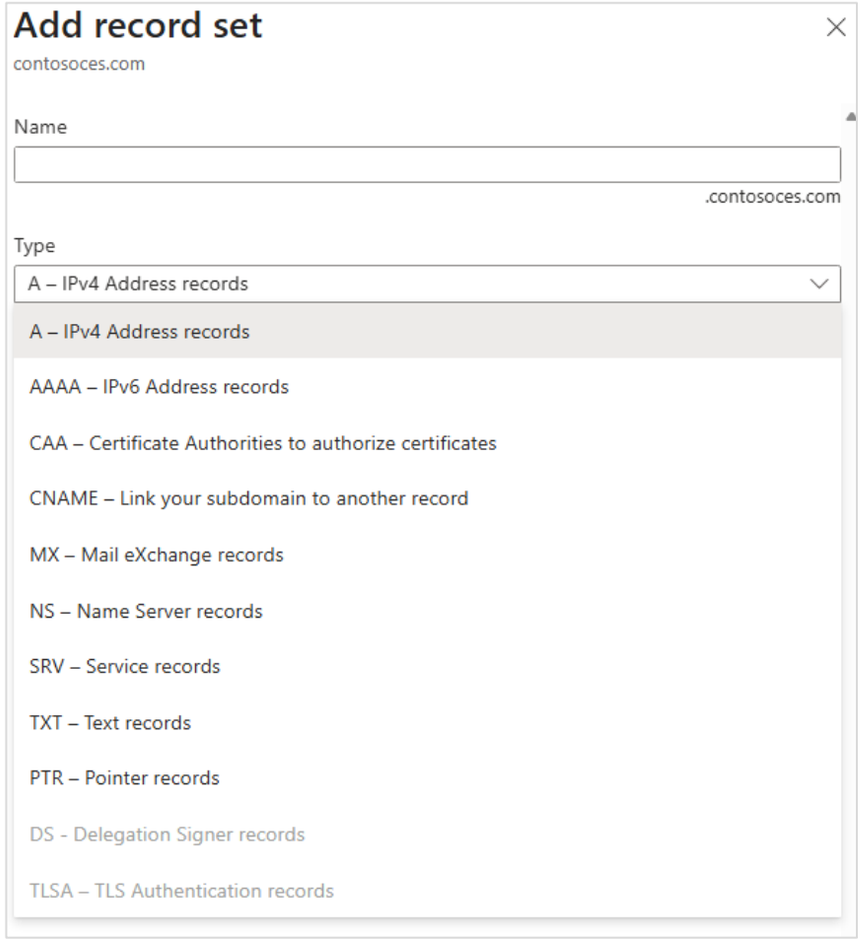
Create a Public DNS Zone and a DNS Record Set

You must claim and register (GoDaddy, Namecheap, or Google Domains) your domain





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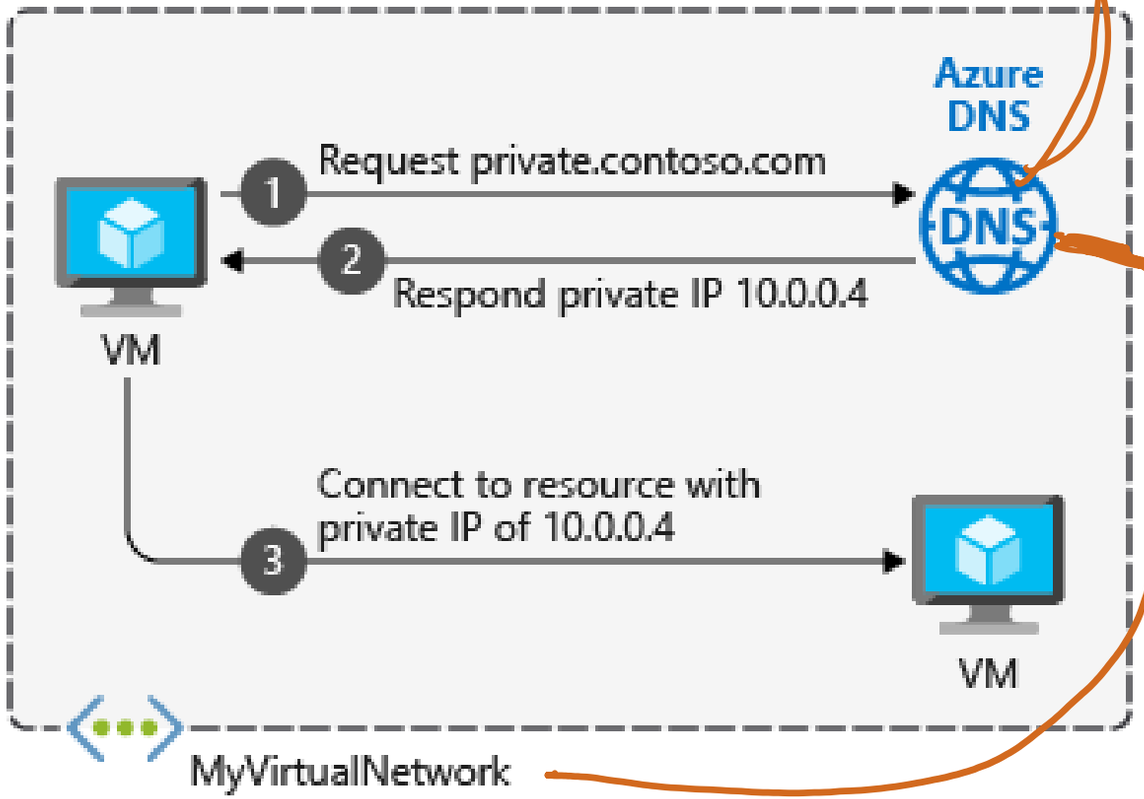
Configure a Private DNS Zone

- Use your own domain names
- Access from any region, any subscription, any VNet, and any tenant
- Provides name resolution for VMs within a VNet and between VNets
- Use auto registration to populate the DNS records for VMs in the zone
- Be sure to create a link from the private zone to the virtual network

Example: Name resolution scoped to a single virtual network



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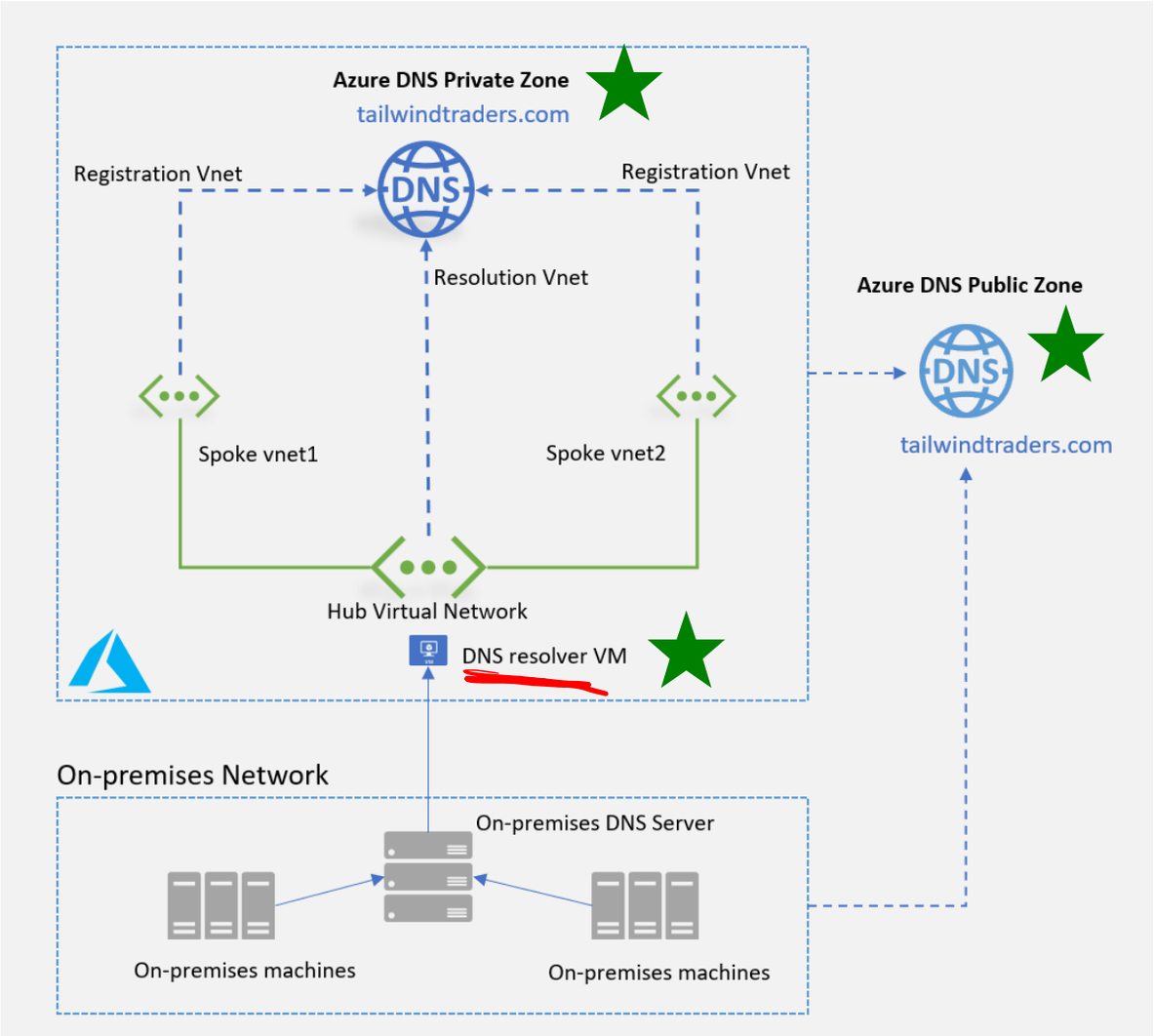
Understand Azure DNS Capabilities

- **Azure Public DNS** is a hosting service that provides name resolution by using Microsoft Azure infrastructure.
- **Azure Private DNS** manages and resolves domain names in the virtual network without the need to configure a custom DNS solution.
- **Azure DNS Private Resolver** resolves DNS queries for private DNS zones from anywhere.



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Lab – Implement Virtual Networks

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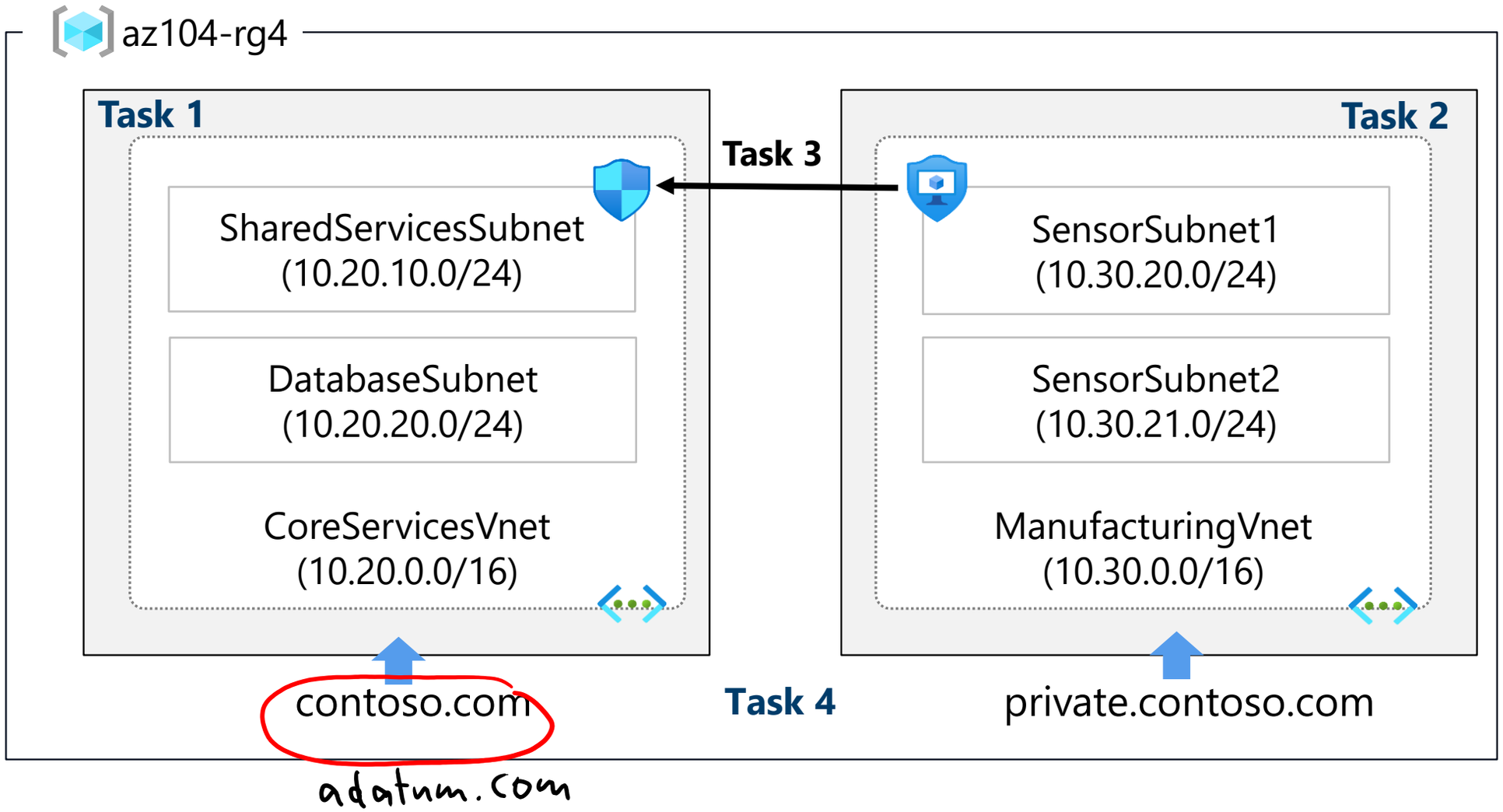
Lab 04 – Architecture diagram



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